

A Finite-Shadow Grammar for Finite Centered Incidence Geometry

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Abstract

This paper records a bounded finite-shadow grammar for finite centered incidence geometry. A row is admitted as a tuple $S = (n, X, I, \rho, \Phi)$, where X is a finite subset of a symmetric group, or a source-locked finite shadow inside one; I is the incidence rule; ρ is the representation or centered-operator channel; and Φ is a replayable fingerprint package. The payoff is not a universal theory: it is a controlled interface that makes the same standard-block and fingerprint tests visible across the published Type-A row, the Fourier/Specht artifact, a signed Type-B/C shadow, and a bounded centered-operator exhibit. The standard-block dictionary is elementary, but it becomes useful here because it is applied on both sides: to the Type-A one-incidence layer and to a signed mirror-union row. The paper includes worked finite evidence showing that Φ is not determined by coarse cycle-type histograms, while also recording why it is not a complete invariant, not a tiling oracle, and not a substitute for the separate theorem routes.

Keywords: finite centered incidence geometry; finite shadows; symmetric groups; Specht fingerprints; centered operators; signed permutations.

MSC 2020: 05A05; 05E10; 05E18; 20B30; 20C30.

1 Introduction and Boundary

Finite centered incidence geometry studies finite incidence-defined rows together with the representation or centered-operator channels that make those rows comparable. The purpose of this paper is to record a bounded grammar for such rows. It is not a final theory of every FCIG branch, and it does not turn the fingerprint channel into a complete invariant.

The paper is built after public source rows. The Type-A standard-space paper supplies the first mature theorem row[Bab26b]. The S_4 finite witness paper supplies a small public finite witness for the structured-subset layer[Bab26c]. The Type-A poset-cone and extension-tiler paper supplies the public chamber/tiler context for rows X01–X02[Bab26e]. The Latin-square and Sudoku operator papers supply public centered-operator sibling context for rows X05–X06[Bab26a, Bab26d]. The present paper does not reopen those proofs; it uses them as source rows or related-work rows in a broader grammar.

The organizing principle is that a row is admitted only after it has a finite ambient set, a stated incidence constraint, a representation or operator channel, and a recorded fingerprint package. Rows without such a finite shadow remain outside the core grammar.

Contributions. The manuscript makes four bounded contributions.

- (i) It fixes a five-slot finite-shadow grammar $S = (n, X, I, \rho, \Phi)$ and an admission table for the current FCIG source rows.
- (ii) It uses the elementary standard-block dictionary as an interface lens: position-value balance is exactly the vanishing of the $(N - 1, 1)$ Specht block, and this lens is applied to both the Type-A row and a signed Type-B/C mirror-union row.

- (iii) It gives worked finite examples from the Type-A and signed Type-B/C artifacts where Φ sees data not determined by coarse cycle-type histograms.
- (iv) It records the guardrails: Φ is descriptive, lattice-qualified where Smith form is used, and never used here as a tiling test or as a substitute for the separate theorem routes.

2 The Finite-Shadow Grammar

Definition 2.1 (finite-shadow grammar row). A finite-shadow grammar row is a tuple

$$S = (n, X, I, \rho, \Phi).$$

Here n records the ambient symmetric degree, $X \subseteq S_n$ is the finite row under study, I is the incidence or selection rule, ρ is the representation or centered-operator channel, and Φ is the recorded fingerprint package.

For signed rows the ambient object may be a source-locked finite shadow $H \leq S_N$, with $X \subseteq H$. The signed Type-B/C row used later has this form with $B_n \leq S_{2n}$, following the signed-shadow conventions used in the companion Type-B/C artifact.

Definition 2.2 (admission levels). The admission levels are governance labels, not new mathematical structures. The active labels are: *core*, *core artifact*, *core exhibit*, *prose*, *pointer only*, *parked*, *future*, and *rejected*. Their role is to prevent an unfinished or overbroad object from entering the manuscript as if it were a closed theorem source.

The full instance matrix is recorded in Section A. The table is part of the grammar: it says not only what enters, but also what remains outside. This is why continuous contact sketches, raw motion rows, the parked exceptional S_6 benchmark, and the separate signed-reversal route are not silently absorbed into the present paper.

Remark 2.3 (the teeth test). A proposed FCIG row fails the teeth test if it lacks any of the five slots. A geometric picture without a finite S_n shadow, a finite set without an incidence rule, or a representation calculation without a source-locked finite object can still be useful research material, but it is not a core row of this paper.

3 The Fingerprint Channel Φ

For $X \subseteq S_N$, the fingerprint channel records block data attached to the Specht/Fourier transforms

$$M_\lambda(X) = \sum_{x \in X} D^\lambda(x),$$

where D^λ is a fixed integral Specht model for the partition $\lambda \vdash N$. The background representation theory is standard[Dia88, JK81].

Definition 3.1 (the Φ -package). The package $\Phi(X)$ may include rational ranks $\text{rank}_{\mathbb{Q}} M_\lambda(X)$, the lattice-qualified invariant $\text{SNF}_\lambda^L(X)$, selected finite-field ranks, block-energy data, transport metadata, and rank mass. Every Smith normal form entry is lattice-qualified; no bare Smith form is used as an invariant independent of the chosen integral model. The use of Smith normal form is standard combinatorial linear algebra[Sta16]; the superscript in SNF^L records that the lattice has been fixed.

The following elementary dictionary is the main reason the grammar is useful rather than decorative. It connects the centered-operator language to a specific Specht block.

Proposition 3.2 (standard-block balance dictionary). *Let $X \subseteq S_N$, and let P_x be the permutation matrix of x on $\mathbb{C}^N$, with rows indexed by values and columns by positions. Put*

$$N_X = \sum_{x \in X} P_x, \quad E_X = N_X - \frac{|X|}{N} J_N,$$

where J_N is the all-ones matrix. Then the following are equivalent:

- (a) X is position-value balanced, i.e. $N_X = (|X|/N)J_N$;
- (b) E_X vanishes on the standard subspace $V_{\text{std}} = \{v : \sum_i v_i = 0\}$;
- (c) the $(N-1, 1)$ Specht block $M_{(N-1,1)}(X)$ vanishes, after choosing the corresponding standard model.

Proof. The permutation representation decomposes as $\mathbb{C}^N = \mathbb{C}\mathbf{1} \oplus V_{\text{std}}$. The constant line records the total row and column sums, while the quotient block is the Specht module of shape $(N-1, 1)$. Subtracting $(|X|/N)J_N$ removes the constant-line part. Thus the centered matrix vanishes exactly when its standard block vanishes. $\square$

Proposition 3.3 (classical left-ideal reading). *Let $d_\lambda = \dim S^\lambda$. For $X \subseteq S_N$, the rank mass used in Φ satisfies*

$$\text{rank_mass}(X) = \sum_{\lambda \vdash N} d_\lambda \text{rank}_{\mathbb{Q}} M_\lambda(X) = \dim_{\mathbb{C}}(\mathbb{C}[S_N]\mathbf{1}_X),$$

where $\mathbf{1}_X = \sum_{x \in X} x \in \mathbb{C}[S_N]$. This is the standard Wedderburn/Fourier interpretation of the left ideal generated by the indicator element [Ser77, Dia88].

Remark 3.4. Definition 3.3 is background, not a novelty claim. Its role is to make the fingerprint channel auditable: when rank mass compresses or fails to compress, the statement is a statement about a classical group-algebra left ideal.

4 Type-A Source Rows

The Type-A source row is anchored by the published standard-space paper [Bab26b]. Its one-incidence layer I_n is the first mature FCIG row in the grammar. The public Type-A poset-cone and extension-tiler paper supplies the separate chamber/tiler context for rows X01–X02 [Bab26e]. This paper uses these definitions and boundaries by citation; it does not merge their theorem claims.

In the notation of Definition 3.2, the Type-A balance theorem says that for even $n \geq 4$, the one-incidence layer has

$$E_{I_n} = N_{I_n} - \frac{|I_n|}{n} J_n = 0,$$

so the standard block $M_{(n-1,1)}(I_n)$ vanishes. This is not merely vocabulary: it is the exact point where the centered-operator language and the Specht block language meet.

The same row can be replayed through the Fourier/Specht fingerprint artifact. The left-ideal checks below are finite support rows, not new counting theorems.

A second Type-A support row shows why Φ cannot be reduced to a conjugacy-class histogram. The Type-A value witness gives two explicit subsets of S_4 , in one-line notation,

$$X_A = \{2314, 2341, 2431, 4231\}, \quad X_B = \{2413, 2431, 4213, 4231\}.$$

They have the same class histogram $(4) : 1, (3, 1) : 2, (2, 1, 1) : 1$, but the $(3, 1)$ block separates them.

Table 1: Type-A sample rows from the Fourier/Specht left-ideal closeout.

row	ambient	$ X $	$\text{rank_mass}(X)$
$I_4(1, 1)$	S_4	8	6
$I_4(0, 2)$	S_4	4	18
$I_5(1, 1)$	S_5	20	41
$I_5(2, 2)$	S_5	8	30

Table 2: Type-A value witness: same class histogram, different Φ -data.

row	$\text{rank}_{\mathbb{Q}} M_{(3,1)}$	$\text{SNF}_{(3,1)}^L$
X_A	2	(1, 2, 0)
X_B	1	(2, 0, 0)

The S_4 witness layer[Bab26c] supplies a smaller finite source row for structured subsets and extension-set comparisons. It is used here as context for how finite rows enter the grammar, not as a license to broaden the Type-A theorem boundary. One notation warning is needed: the public poset-cone paper uses a label P038 for a proved extension-tiler witness, while row X10 in the instance matrix refers to a parked exceptional S_6 benchmark. These are distinct source labels; this paper cites the public tiler witness as chamber context and keeps the exceptional benchmark parked.

5 Signed Type-B/C Source Rows

The signed source row comes from the companion signed Type-B/C dependency artifact. It works with the finite shadow $B_n \leq S_{2n}$, using the pair model for signed coordinates. Standard Coxeter background for signed permutations is classical[BB05].

The signed incidence rows $BI_n(k, l)$ record positive fixed signed pairs and sign-flip-in-place pairs. The signed artifact verifies that the same fingerprint channel can process these rows through the ambient symmetric group. The manuscript uses this as a bounded second finite-shadow instance.

Table 3: Signed finite-shadow rows recorded in the signed replay.

row	ambient	$ X $	$\text{rank_mass}(X)$
$BI_3(0, 0)$	S_6	16	90
$BI_3(0, 1)$	S_6	12	300
$BI_3(1, 0)$	S_6	12	300
hash-ordered size- $ B_3 $ control	S_6	48	715
$BI_4(0, 1)$	S_8	64	10710
$BI_4(1, 1)$	S_8	48	17220

The fourth row of Table 3 is not an unspecified random draw: it is the deterministic hash-ordered size- B_3 control recorded by the signed artifact. Its role is only to show that the signed rows are not being compared against a moving baseline.

The first place where Definition 3.2 genuinely crosses type is the signed mirror-union. Put

$$Y_n^B = BI_n(1, 0) \sqcup BI_n(0, 1) \subseteq B_n \leq S_{2n}.$$

Let D_m denote the number of derangements of m letters.

Proposition 5.1 (signed mirror-union balance source row). *For $n \geq 3$, the signed mirror-union satisfies*

$$|Y_n^B| = n2^n D_{n-1}, \quad N_{Y_n^B} = 2^{n-1} D_{n-1} J_{2n}.$$

Equivalently, the ambient standard block vanishes:

$$M_{(2n-1,1)}(Y_n^B) = 0.$$

Source-locked proof sketch. The signed source lock decomposes the ambient permutation module $\mathbb{C}^{2^n}$ into the constant line plus two nontrivial signed constituents: a pair-standard channel and a reflection channel. For a single layer $BI_n(k, l)$, the pair-standard scalar is proportional to $k+l-1$, while the reflection scalar is proportional to $k-l$. Thus a single nonempty layer cannot kill both channels. The two layers $BI_n(1, 0)$ and $BI_n(0, 1)$ each kill the pair-standard channel, and their equal sizes make the reflection scalars cancel in the union. Definition 3.2 then gives the displayed position-value balance and the vanishing of $M_{(2n-1,1)}$. $\square$

For example, at $n = 3, 4, 5, 6$ the signed-balance check gives $|Y_n^B| = 24, 128, 1440, 16896$ and constants $4, 16, 144, 1408$, respectively. This is the concrete signed use of the standard-block dictionary: Type A balances a single one-incidence layer in its mature theorem row, while the signed row balances a mirror-union because the ambient standard splits into two signed channels.

The row $BI_3(0, 0)$ compresses to the sign-subgroup index background. The mixed rows, such as $BI_3(0, 1)$ and $BI_4(0, 1)$, are the bounded signal for this paper: they are neither random-sized full support nor a promoted closed formula. They show that the finite-shadow interface sees signed incidence structure while keeping the signed artifact as a bounded dependency.

The signed value witness gives the signed analogue of the Type-A value witness. In one-line notation on S_6 ,

$$X_A = \{123465, 124365\}, \quad X_B = \{123465, 214356\}.$$

Equivalently, inside B_3 , these are $X_A = s_3 \langle s_2 \rangle$ and $X_B = s_3 \langle s_1 s_2 s_3 \rangle$. They have the same signed-cycle histogram and the same rank mass, but the standard block over S_6 separates them.

Table 4: Signed value witness: same signed histogram, different standard block.

row	rank_mass	rank $_{\mathbb{Q}} M_{(5,1)}$	SNF $_{(5,1)}^L$
X_A	360	4	(1, 2, 2, 2, 0)
X_B	360	2	(1, 1, 0, 0, 0)

No broader Coxeter-group statement is made in this section. The signed row is a source-locked finite artifact and a grammar test.

6 A/B/D Grammar Contrast

The A/B/D material enters this paper as grammar contrast. Its purpose is to show how the same admission discipline reacts when the signed setting changes the row shape. The signed-shadow routing table and signed-balance notes are the source locks for this contrast.

The contrast is deliberately asymmetric. The Type-A row has a published theorem source, the Type-B/C row has a closed bounded artifact plus the signed mirror-union source row of Definition 5.1, and the D-side signed reversal material belongs to the separate signed-reversal route. This paper may point to that route, but it does not import theorem statements from it.

In this sense the grammar has teeth: it can admit a bounded signed finite shadow while refusing to smuggle in a neighboring theorem route whose source locks are not yet closed. It also shows a real change of behavior. The Type-A row balances through one centered channel and is parity-fragile on the odd side; the signed row must balance two channels at once, but the mirror-union does so for every $n \geq 3$.

Table 5: How the same grammar test behaves across the A/B/D sources.

feature	Type-A row	signed Type-B/C row
ambient standard	one standard channel V_{std}	ambient standard splits into pair-standard and reflection channels
balanced object	single layer $I_n(1, 1)$ in the mature Type-A theorem row	mirror-union $BI_n(1, 0) \sqcup BI_n(0, 1)$, not a single signed layer
parity behavior	all-even theorem; odd side kept source-bounded, with $n = 5$ exceptional and $n = 7$ already unbalanced	parity-free: the mirror-union is balanced for every $n \geq 3$
what the grammar shows	centered operator balance and Specht $(n - 1, 1)$ vanishing are the same test	the same test survives, but the signed shadow forces cancellation across the two mirror layers
what remains outside this paper	odd and shifted variants beyond their locked evidence	D-side theorem-level material remains pointer-only

7 Value and Limits of Φ

The value of Φ is operational. In the source-locked audits, it records finite structure not visible from coarser summaries alone. The Type-A and signed Type-B/C artifacts both provide finite evidence that this channel is useful for comparison.

Table 6: What the worked witnesses show.

source	finite observation	allowed conclusion
Type-A Fourier witness	same class histogram; ranks 2 and 1 at $(3, 1)$	Φ is not determined by class histograms
signed Type-B/C witness	same signed-cycle histogram; ranks 4 and 2 at $(5, 1)$	the same value test survives in a signed finite shadow
signed mixed layers	$BI_3(0, 1)$ has rank mass 300, below the size- B_3 hash-ordered control 715	signed incidence rows carry visible finite structure
M19/X05 exhibit	blind rank-locus mining repeatedly recovers the same order-72 group G_0 when the full coset occurs	Φ can recover a hidden group from centered-operator data

7.1 A controlled X05 exhibit: M19

The X05 Sudoku/Latin row gives a separate controlled example of the same point. For an order-nine square L , take $X = S_9$ acting by symbol relabeling and let

$$E_{\gamma, L}(i, j) = \gamma(L(i, j)) - 5$$

be the centered operator, restricted to the standard subspace. The fingerprint package records the rank-drop locus

$$\Gamma_{\text{rank}}(L) = \{\gamma \in S_9 : \text{rank } E_{\gamma, L} < 8\}$$

together with the hidden stabilizer recovered by blind coset mining. This is the same grammar slot Φ , now applied to a centered-operator sibling of the Type-A rows.

The published Sudoku operator paper isolates the original M19 rank-locus phenomenon; see [Bab26d]. The centered Latin-square paper supplies the sibling determinant/rank-obstruction context for the Latin side of the centered-operator row[Bab26a]. The FCIG census source then asks whether M19 is isolated. Its durable, source-locked answer is that the large order-72 hidden coset is not a one-off. Whenever that full coset occurs in the corrected census, the hidden group is the same fixed subgroup $G_0 \leq S_9$, the affine symmetry group of the natural $1, \dots, 9$ value grid. In the corrected 1000+1000 census, this full G_0 -coset appears in both Sudoku and Latin samples. The visible magic-band refinement is observed only on the Sudoku side. Complement relabeling is the baseline symmetry of the centered rank-drop condition.

Table 7: M19/X05 census facts used here.

item	source-locked status
full hidden coset	order 72; recurring signal, not a single M19 accident
recovered group	same fixed $G_0 \leq S_9$ whenever the full coset occurs
corrected census	1000 Sudoku samples and 1000 Latin samples inspected in the source package
magic-band refinement	observed on the Sudoku side only
open exception	order-32 <code>lat565</code> remains uncharacterized

This exhibit is useful because Φ recovers a recurring finite group from the data without being handed the group in advance. It is not used here as a theorem pillar. The frequencies are empirical, the mechanism selecting the arrays is open, and the out-of- G_0 population, including `lat565`, remains uncharacterized.

The same audits also fix the limit. The fingerprint is not a complete invariant, and rank data is not used as a decision oracle for unrelated structural questions. This section records the value and the guardrails together.

8 Boundaries and Future Routes

The grammar leaves several routes outside the current claim boundary. The signed-reversal program is a separate theorem route. Modular centered operators, motion/contact rows, non-convex Coxeter-pure rows, the parked exceptional S_6 benchmark, and the blocked numerical bridge remain future or parked work unless their own gates promote them.

The X05/M19 exhibit has its own boundary. This paper may use it to show that Φ can recover a recurring hidden group in a centered-operator rank locus. This paper does not claim a general Sudoku theorem, a full mechanism theorem for the rank drop, or an explanation of the heterogeneous out-of- G_0 exceptions such as `lat565`.

The shifted-diagonal material and the 5568 bridge candidate remain fenced. The shifted-diagonal row is useful as a balance-control exhibit, but the all- n statement is still a proof obligation. The numerical bridge involving 5568 is not used as a theorem or as evidence of an identified morphism.

This boundary is part of the result of the interface. The grammar is useful only if it admits source-locked finite rows while refusing rows whose finite shadow, representation channel, or fingerprint package is not yet defined. If later drafting removes the worked examples in Tables 1 to 8, the honest fallback is a shorter interface note rather than a full paper.

A Source and Instance Manifest

This appendix records the finite-shadow instance matrix used by the paper. Public scholarly rows are cited in the bibliography. Companion artifact rows are included only as bounded

Table 8: Future routes and open debts kept outside the present claim boundary.

route or debt	current role	admission condition before promotion
signed-reversal route	separate theorem-level route	must close its own remaining gates; kept pointer-only here
P16 modular centered operators	future modular route	needs finite-shadow channel and modular proof boundary
future grammar extensions	future grammar branches	need source locks and non-overlap with this paper’s claims
M19 and <code>lat565</code>	controlled X05 exhibit	mechanism and heterogeneous exceptions must be characterized separately
shifted-diagonal row	balance-control exhibit	all- n statement remains a proof obligation
5568 bridge candidate	blocked numerical coincidence/bridge candidate	requires an identified morphism or explicit non-bridge closeout
exceptional S_6 benchmark	parked benchmark	no theorem import without its own prior-art and source-lock gates

source rows for the interface; they are not cited as independent mathematical authority unless separately released.

Table A.1. Finite-shadow instance matrix.

ID	row	admission	role and boundary
X01	Type-A poset cones $L(P)$	core/prose	Type-A chamber context; no new extension-set theorem.
X02	Type-A extension tilers	prose/exhibit	Exact-cover context; no Φ -tiling test.
X03	subgroup/coset rows	core exhibit	Controls for rank mass and left-ideal background.
X04	finite S_4 witness	core exhibit	Published bounded witness only.
X05	Sudoku/M19 rank loci	core/prose	Controlled M19 exhibit; <code>lat565</code> open.
X06	Latin determinant-defect loci	prose	Centered-operator sibling; no proof import.
X07	magic/endpoint defect loci	future	No unresolved finite-field theorem promoted.
X08	one-incidence layers $I_n(k, l)$	core	Type-A theorem boundary; no 5568 bridge.
X09	prefix/shadow subsets	prose	Folded into finite-witness provenance.
X10	exceptional S_6 benchmark	parked	Parked benchmark only; distinct from the public poset-paper P038 tiler.
X11	nonconvex Coxeter-pure tilers	future	Needs its own finite-shadow source lock.
X12	motion/contact atlas objects	rejected/future	Fails current finite-shadow teeth test.
X13	Fourier/Specht fingerprints	core artifact	Descriptive Φ , SNF^L , ranks, rank mass.
X14	grammar interface	prose/meta	Organizes rows; not a theorem by itself.
X15	signed Type-B/C shadows $B_n \leq S_{2n}$	core artifact	Bounded signed source; no broader Coxeter claim.
X16	modular centered operators	future	Reserved for later modular route.
X17	signed reversal noncentral layer	pointer only	Separate signed-reversal route only; no theorem import here.

Public and companion pointers. The public Type-A standard-space paper, the public S_4 finite witness paper, the public Type-A poset-cone/extension-tiler paper, and the public Latin-square and Sudoku operator papers are scholarly pointers for rows X01, X02, X04, X05, X06, X08, and related prose rows. The label P038 is overloaded across the wider source archive: the public Type-A poset-cone paper contains a proved extension-tiler witness with that label, while row X10 above is a parked exceptional benchmark and imports no theorem here. The Fourier/Specht and signed Type-B/C rows are companion dependency artifacts for the present grammar paper. The M19 census package is a companion source-lock pointer for the X05 exhibit and is not used as public mathematical authority unless separately exported.

B Reproducibility Notes

This repository contains the LaTeX source, compiled title-named PDF, public boundary notes, source manifest, and SHA-256 manifest for the paper snapshot. The paper is an interface paper: its finite numerical rows are replayable in the cited public companion repositories or in bounded companion artifacts summarized by Section A. No theorem in this paper depends on treating those companion artifacts as public mathematical authority.

The paper can be rebuilt from `paper/main.tex` with the standard sequence

`pdflatex, bibtex, pdflatex, pdflatex.`

The expected output is

`paper/A_Finite-Shadow_Grammar_for_Finite_Centered_Incidence_Geometry.pdf.`

Data and Code Availability

The companion repository contains the LaTeX source, compiled PDF, source manifest, claim-boundary notes, and SHA-256 manifest for this paper snapshot. The finite computations cited here are either in the cited public companion repositories or recorded as source-locked dependency artifacts summarized by the source manifest; this repository does not claim a standalone replay engine for every upstream row.

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